



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

A PROJECT TITLE

ON

Customer Lifetime Value Prediction Using Elastic Net Regression

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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CSE DEPARTMENT





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DEPARTMENT OF __CSE_____

Name of Title: Customer Lifetime Value Prediction Using Elastic Net Regression

1. Abstract:

Customer Lifetime Value (CLV) prediction is a crucial application of machine learning in the e-commerce industry, enabling businesses to estimate the total revenue a customer is likely to generate throughout their relationship with a company. Accurate CLV prediction helps organizations identify high-value customers, improve customer retention, optimize marketing campaigns, and maximize long-term profitability. However, customer data often contains highly correlated features, making traditional regression models less effective due to issues such as multicollinearity and overfitting. To overcome these challenges, this project employs Elastic Net Regression, a regularized machine learning algorithm that combines the advantages of Lasso (L1) and Ridge (L2) regression techniques.

The proposed model uses historical customer transaction data, including purchase frequency, recency, monetary value, average order value, browsing behavior, and demographic information. Before training the model, the dataset undergoes preprocessing steps such as handling missing values, encoding categorical variables, feature scaling, and data cleaning to improve prediction accuracy. Elastic Net Regression performs both feature selection and coefficient regularization, enabling the model to produce stable and reliable predictions while minimizing overfitting.

The predicted Customer Lifetime Value allows businesses to segment customers based on their future revenue potential and develop personalized marketing strategies, loyalty programs, targeted promotions, and retention campaigns. This approach supports efficient resource allocation and enhances customer relationship management. The performance of the model is evaluated using standard regression metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2) score, ensuring the reliability of the predictions. Overall, this project demonstrates the effectiveness of Elastic Net Regression in predicting Customer Lifetime Value for e-commerce businesses. By accurately estimating customer value, organizations can make data-driven decisions, improve customer satisfaction, increase marketing return on investment (ROI), and achieve sustainable business growth while maintaining a competitive advantage in the rapidly evolving digital marketplace.

2. INTRODUCTION:

The rapid growth of e-commerce has transformed the way businesses interact with customers. Online retailers collect large volumes of customer data through purchases, website interactions, product searches, payment history, and customer feedback. Analyzing this data allows businesses to understand customer behavior and make informed business decisions. One of the most significant business metrics derived from customer data is Customer Lifetime Value (CLV).

Customer Lifetime Value refers to the estimated total revenue that a customer will contribute to a business during their entire relationship with the company. Predicting CLV enables organizations to identify valuable customers, reduce customer acquisition costs, improve customer retention, and design personalized marketing strategies. Companies with accurate CLV prediction models can focus their resources on customers who are likely to generate higher long-term profits.

However, predicting CLV is a complex regression problem because customer behavior depends on multiple correlated variables such as purchase frequency, recency, spending habits, discount usage, browsing history, and demographic information. Traditional linear regression models often suffer from multicollinearity and overfitting when handling such datasets, reducing prediction accuracy.

To overcome these limitations, this project implements Elastic Net Regression, a regularized machine learning algorithm that combines the advantages of Ridge Regression and Lasso Regression. Ridge Regression reduces overfitting by shrinking regression coefficients, while Lasso Regression performs feature selection by eliminating less important features. Elastic Net combines both penalties to achieve better prediction performance, especially for datasets with highly correlated variables.

The project follows a structured workflow involving data collection, preprocessing, feature engineering, model training, hyperparameter tuning, prediction, and performance evaluation. Standard evaluation metrics such as MAE, MSE, RMSE, and R^2 Score are used to assess the effectiveness of the model.

The developed system assists e-commerce businesses in identifying high-value customers, optimizing customer relationship management (CRM), improving marketing efficiency, increasing customer retention, and maximizing overall business revenue.

3. SOFTWARE REQUIREMENTS:

Operating System

- Windows 10/11, Linux, or macOS

Programming Language

- Python 3.10 or above

Development Environment

- Jupyter Notebook / Google Colab / VS Code

Libraries

- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Joblib (for model saving)

Dataset

- Customer transaction dataset (CSV format)

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: At least 2 GB free space

Software Tools

- Anaconda (Optional)
- Git (Optional)
- Web Browser (Chrome/Edge/Firefox)

Machine Learning Algorithm

- Elastic Net Regression

Evaluation Metrics

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R-Squared (R^2) Score